

A Robust Position Approach Based on Masked KalmanNet for GNSS/LEO/INS Integrated Navigation System

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Abstract—In the domains of the Internet of Things (IoT) and intelligent transportation, the importance of integrity monitoring has become increasingly prominent with the rapid development of autonomous driving. In complex environments, the redundancy of effective observations often fails to meet the stringent requirements of integrity monitoring. Moreover, due to the effects of multipath (MP) and non-line-of-sight (NLOS), receiver noise frequently deviates from the Gaussian distribution, thereby significantly degrading the performance of traditional global navigation satellite system (GNSS)/inertial navigation system (INS) integrated navigation algorithms. Capitalizing on the rapid deployment of Low Earth Orbit (LEO) satellite constellations, this article proposes a novel integrity monitoring algorithm for GNSS/LEO/INS tightly coupled (TC) systems within the KalmanNet framework. First, LEO satellites provide enhanced observational redundancy, thereby improving the reliability of integrity monitoring. Second, the proposed algorithm relaxes the Gaussian noise assumption and instead utilizes a masked convolutional neural network (CNN)–long short-term memory (LSTM)–attention (CLA) network to learn features from the observation data. Experimental results demonstrate that the proposed algorithm achieves root-mean-squared error (RMSE) reductions of 55.53%, 51.74%, 36.42%, and 29.50% compared to standard point positioning (SPP), TC IGG3, RLAKF, and BSRMCL algorithms, respectively.

Index Terms—Integrity monitoring, KalmanNet, low Earth orbit (LEO) satellite enhancement, masked matrix, protection levels (PLs).

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I. INTRODUCTION

THE Internet of Things (IoT) technology is closely related to human life. Global navigation satellite system (GNSS), as an important component of the IoT system, plays an irreplaceable role in the field of intelligent transportation, and its integrity is receiving increasing attention [1], [2]. GNSS integrity originates from civil aviation. In the field of intelligent transportation, with the continuous advancement of autonomous driving technology, the navigation integrity of vehicles has received increasing attention [3], [4]. In many practical IoT navigation applications, due to the influence of multipath (MP) and nonline-of-sight (NLOS), the observation noise does not completely follow a Gaussian distribution [5], [6]. This makes it difficult for traditional state-space modeling to accurately present noise information [7], [8]. In recent years, with the construction of a low Earth orbit (LEO) satellite constellation, research on enhancing GNSS navigation with LEO satellites has gradually become a hot topic [9], [10], [11], [12], [13]. As shown in Fig. 1, the increase in measurements can alleviate the problem of GNSS signal unavailability in complex environments. Furthermore, it is also significant for fault detection and elimination (FDE), because more observation data provide greater degrees of freedom, enabling the system to detect smaller minimum detectable deviations (MDBs). Due to the lack of downlink observation data from LEO satellites, this article first provides detailed LEO downlink observation data from satellite tool kit (STK), taking into account various errors to make it as close to real observation data as possible.

A. Related Work

In response to the nonlinear and non-Gaussian characteristics encountered by navigation filters, deep neural networks (DNNs) have demonstrated significant empirical success in practical applications, offering innovative solutions to these challenges [14], [15], [16]. The convolutional neural network (CNN) excels in processing spatial data, such as images, by efficiently capturing local patterns (e.g., edges and textures) within grid structures through local perception, weight sharing, and hierarchical feature extraction. CNN has been widely used in tasks like image classification and object detection, while the long short-term memory (LSTM) network specializes in

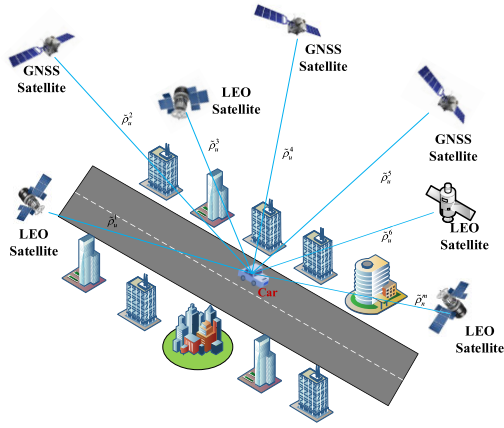


Fig. 1. Multisource navigation in future LEO enhanced scenarios.

time-series data, such as speech and text. It addresses long-term dependency problems through gating mechanisms (input gates, forget gates, and output gates) and cell states, adapting to variable-length sequence inputs. LSTM performs well in prediction and generation tasks. When combined, CNN and LSTM are well suited to handle spatiotemporal mixed data. These data-driven models have proven effective in capturing the subtleties of complex processes, eliminating the need for explicit domain descriptions [17]. Consequently, an alternative strategy for achieving state estimation involves the deployment of deep learning techniques to directly extract noise characteristics from raw data, thereby bypassing the requirement for explicit and accurate prior knowledge of the state-space model. DNNs, including recurrent neural networks (RNNs), gated recurrent units (GRUs), and attention mechanisms, have shown great promise in time-series tasks, especially in challenging environments. By training these networks end-to-end and in a model-agnostic manner using large datasets, they offer robust performance. However, DNNs cannot integrate domain knowledge in a principled way, unlike structured state-space models. Although DNNs are making significant breakthroughs in the field of navigation, these data-driven approaches are often criticized for their lack of interpretability.

Recent research has increasingly focused on applying deep learning techniques to solve GNSS positioning challenges. Su et al. [18] propose a deep learning-assisted multimodal measurement compensation method for GNSS/inertial navigation system (INS) loosely coupled (LC), which effectively enhances positioning accuracy in complex environments. Jing et al. [19] present a real-time GNSS deception detection algorithm utilizing feature processing. With detection times of less than 10 μ s, this approach significantly improves the $F1$ score, offering a novel intelligent solution for GNSS deception detection. Jiang et al. [20] introduce a voting-based GNSS/INS tightly coupled (TC) algorithm, which effectively improves 3-D positioning accuracy in complex urban environments. In addition, Jiang et al. [21] propose a deep learning-based method for detecting outliers in GNSS observations from smartphones. Experimental results demonstrate that this algorithm can effectively detect large outliers, enhancing the stability of positioning results. Hu et al. [22] present a new

GNSS positioning algorithm based on backpropagation neural networks. Three distinct network structures are proposed, with experimental results showing that the neural network method significantly reduces positioning errors compared to conventional algorithms. Deep learning methods have also found numerous applications in integrated navigation, particularly in challenging environments with limited satellite visibility. Deep learning can be utilized to predict and correct the divergence of inertial navigation, thereby reducing positioning errors in scenarios where GNSS signals are unavailable [23], [24], [25]. In integrated navigation systems, non-Gaussian observation noise and nonlinear dynamics often lead to performance degradation in traditional Kalman filter (KF). To address these challenges, researchers have introduced KalmanNet and demonstrated its effectiveness through LC methods [14], [15], [26]. For feature extraction of GNSS observation data, the CNN-LSTM-attention (CLA) framework has been widely employed [17], [27], [28], [29], [30]. However, in GNSS/INS TC architecture, this approach becomes less effective due to the temporal variation of observation values.

B. Motivations and Contributions

To adapt the KalmanNet framework to the TC model, this article proposes incorporating the CLA algorithm with the assistance of a masked matrix. First, the maximum sequence length is determined based on the maximum number of observed satellites, with zero-padding applied to the sequence for data alignment. In order to avoid the CLA network learning zero-padding elements, inspired by the Transformer model [31], this article introduces a masked matrix. This logical masked matrix distinguishes zero-padding elements from actual data, ensuring that the network is not influenced by the zero-padding. The proposed algorithm combines both model- and data-driven approaches, offering strong interpretability while avoiding the assumption of Gaussian distribution for noisy models. In addition, the FDE is a critical component of integrity monitoring, and an increasing number of scholars have made significant contributions to this field. In earlier studies, hypothesis testing and parameter estimation are often treated as independent processes [32]. However, in practical applications, there is a clear dependency between the two. Ignoring this relationship can lead to overly optimistic integrity risk assessments. Ciuban et al. [33], [34] thoroughly analyzed the relationship between parameter estimation and hypothesis testing, proposing optimization methods for applications such as autonomous driving. This article integrates fault detection with parameter estimation to effectively detect and eliminate observations containing faults, ensuring the reliability and robustness of positioning results.

The main contributions of this article are given as follows.

- 1) This article presents a detailed process for simulating LEO downlink data, considering various error factors and aiming to closely approximate real observation data.
- 2) This article denotes a CLA network assisted with a masked matrix, which realizes the GNSS/INS/LEO TC algorithm under the KalmanNet structure. In addition, it eliminates the need for matrix inversion, improving real-time computational efficiency.

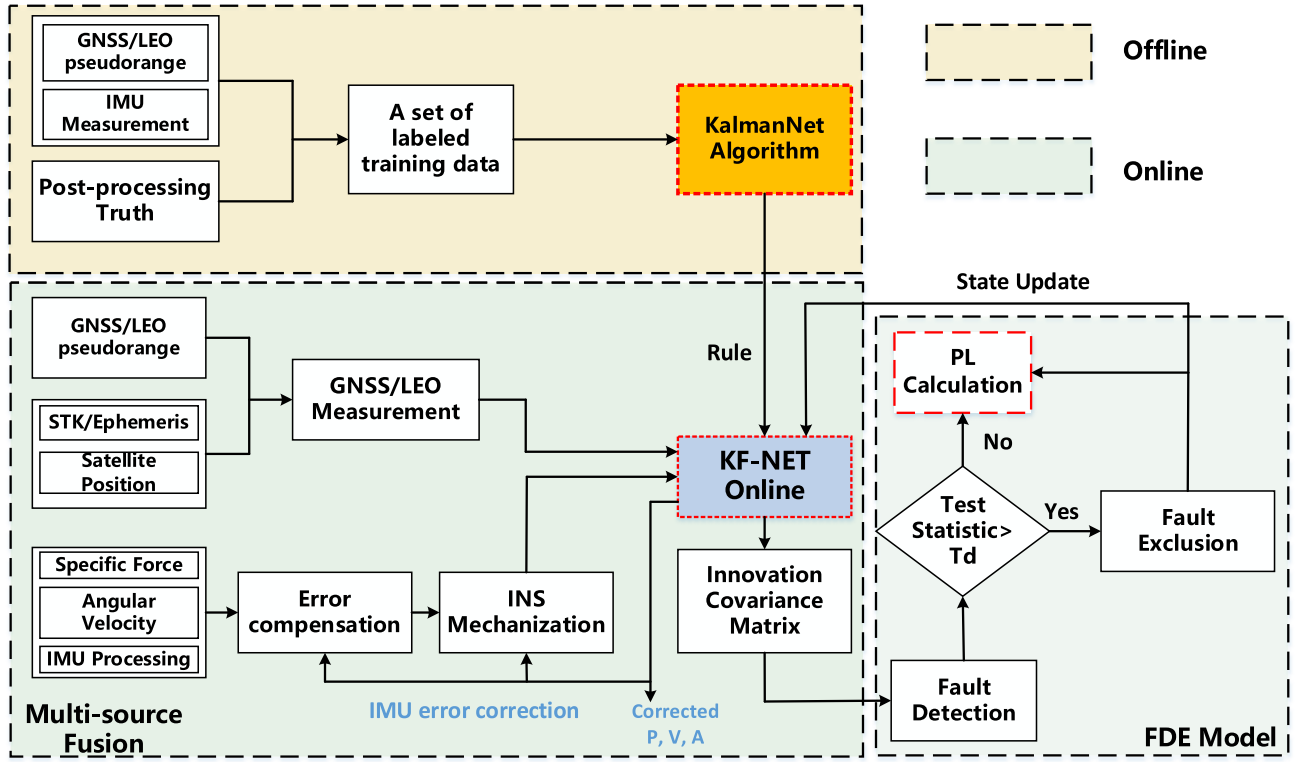


Fig. 2. Proposed GNSS/INS/LEO TC framework in this article.

- 3) This article explores the relationship between parameter estimation and hypothesis testing, integrating FDE with parameter estimation to enhance integrity monitoring performance.
- 4) This article uses two real-world dynamic observation datasets, divided into training and testing datasets, and incorporates simulated LEO downlink data. Through a semi-physical simulation experiment, it demonstrates that the proposed algorithm effectively improves the reliability and robustness of positioning results in complex environments. Compared to standard point positioning (SPP), TC IGG3, RLAKF, and BSRMCL, our algorithm significantly reduces the root-mean-squared error (RMSE) of positioning error.

The remaining sections of this article are given as follows. Section II provides detailed steps for simulating LEO satellite downlink data and presents the overall framework of the algorithm. Section III provides a detailed experimental scheme and results, and the experimental results are discussed. Section IV is the conclusion.

II. ALGORITHM FRAMEWORK

In this section, we introduce the overall structure of the proposed algorithm, as illustrated in Fig. 2. The algorithm consists of two main modules: offline training and online testing. The purpose of the offline training module is to learn and store network parameters, which are used in the online testing model for real-time navigation computation. In online testing, the FDE model is used for integrity monitoring. If a fault is detected, the algorithm will perform fault elimination,

update the state parameters accordingly, and reoutput the corrected navigation solution.

A. Simulation of LEO Satellites Observations

Due to the current lack of actual downlink observation data from LEO satellites, LEO satellite orbits are generated through STK simulation using the high-precision orbit propagator (HPOP) model. The modeling of pseudorange measurements for LEO satellites is given as follows [35]:

$$\tilde{\rho}^i = \rho^i + c\delta t_u - c\delta t^i + I_u^i + T_u^i + \epsilon^i, \quad i = 1, \dots, m \quad (1)$$

where i indicates the index of the i th LEO satellite. The speed of light is represented by c . δt_u is the offset of the receiver clock. δt^i is the offset of the satellite clock. I_u^i and T_u^i represent the delay due to the ionosphere and troposphere, respectively. ϵ^i is the sum of errors caused by MP/NLOS and receiver noise. m represents the number of visible LEO satellites. Tropospheric delay and satellite receiver clock errors can be mitigated using the same techniques employed in traditional GNSS. The ionospheric altitude is generally between 100 and 1000 km, but some LEO satellites are located in the middle of the ionosphere, so ionospheric delay modeling cannot be done using the same model as GNSS. Adopting the following equidistant weighting in ionospheric modeling in this article:

$$I_u^i = \begin{cases} I_{Klo}, & h > h_H \\ \frac{h - h_L}{h_H - h_L} I_{Klo}, & h_L \leq h \leq h_H \\ I_{Klo}, & h < h_L \end{cases} \quad (2)$$

where h_H and h_L are the upper and lower ionospheric heights, respectively. As illustrated in Fig. 3, ionospheric modeling for

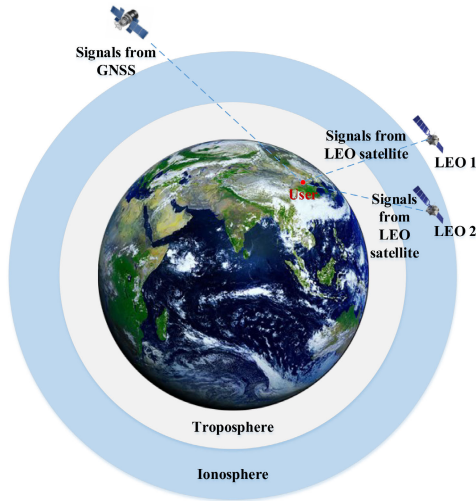


Fig. 3. Ionosphere conditions of satellites at different orbital altitudes.

LEO satellites is determined by their orbital altitude. When the altitude of an LEO satellite exceeds the upper limit of the ionosphere, the Klobuchar model I_{Klo} is employed for ionospheric delay correction, consistent with traditional GNSS [36]. However, when the LEO satellites operate within the ionospheric layer, the equidistant weighting model is applied instead. Nevertheless, due to the inherently nonuniform spatial distribution of ionospheric electron density, the equidistant weighting model may yield an optimistic estimation of the ionospheric delay.

Pseudorange measurement error is the core indicator of positioning accuracy, which is influenced by three aspects: satellite, propagation, and receiver. The variance $\sigma_{\tilde{e}}^2$ measured by each satellite depends on several factors. As illustrated in Fig. 4, signals transmitted from satellites are subject to varying degrees of distortion and degradation due to several error sources, including ionospheric and tropospheric delays, MP/NLOS effects, and receiver noise. Therefore, for each satellite

$$\sigma_{\tilde{e}}^2 = \sigma_{\tilde{e},URA}^2 + \sigma_{\tilde{e},iono}^2 + \sigma_{\tilde{e},tropo}^2 + \sigma_{\tilde{e},mp}^2 + \sigma_{\tilde{e},rcvn}^2 \quad (3)$$

where $\sigma_{\tilde{e},URA}^2$ is the variance of user range accuracy (URA). $\sigma_{\tilde{e},rcvn}^2$ is the variance of receiver noise error. $\sigma_{\tilde{e},iono}^2$, $\sigma_{\tilde{e},tropo}^2$, and $\sigma_{\tilde{e},mp}^2$ represent the variances of the tropospheric, ionosphere, and MP/NLOS, respectively.

The modeling method in [35] can be used for URA, tropospheric error, and ionospheric error. However, for MP effects, this article adopts the following method. As shown in Fig. 5, the relationship between carrier-to-noise ratio (C/N0), satellite elevation, and MP/NLOS error is examined using GNSS observational data collected from vehicles located in Caidian, Wuhan. The analysis reveals that the relationship between MP/NLOS error and both C/N0 and elevation is nonlinear and intricate, making it difficult to derive a precise analytical model. This article refers to [37] and provides the relationship between the variance caused by MP/NLOS, the satellite observation elevation, and C/N0. In addition, the relationship between the observation elevation of

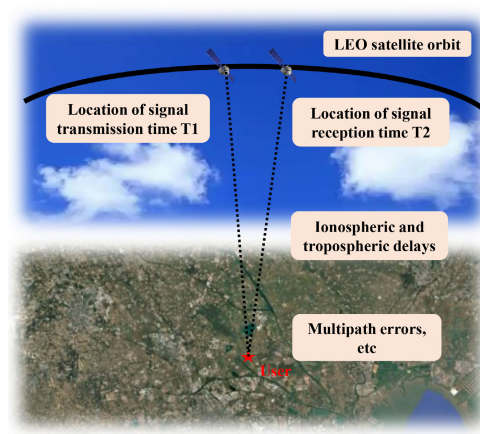


Fig. 4. Analysis of the main error sources considered in this article.

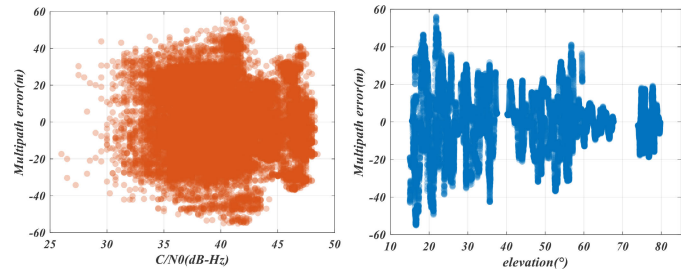


Fig. 5. Relationship between MP/NLOS error, satellite elevation, and C/N0.

LEO satellites and C/N0 is designed according to [35]

$$\sigma_{\tilde{e},mp}^2 = \begin{cases} \frac{\sin^2 e}{\left(10^{-\frac{CN0-s1}{a}} \left(\left(\frac{A}{10^{-\frac{s0-s1}{a}}} - 1 \right) \frac{CN0-s1}{s0-s1} + 1 \right) \right)}, & CN0 < s1 \\ 1, & CN0 \geq s1 \end{cases} \quad (4)$$

the meanings of A , e , $s0$, and $s1$ can all be found in [35]. Unlike GNSS position calculation, the LEO positioning simulated in this article is actually the inverse process of GNSS positioning. Due to the higher speed and the shorter transit time of LEO satellites compared to GNSS satellites, the motion of LEO satellites during the signal propagation period cannot be ignored. As shown in Fig. 4, assuming the signal reception epoch T2 and the signal transmission epoch T1, we can obtain the pseudorange observation at epoch T2. However, we ignore the error caused by the motion of the LEO satellite at this time. This article also deals with this error as follows.

- 1) The distance between the satellite and the receiver is ρ_{T2} at epoch T2, and the signal propagation time can be calculated using the following formula:

$$\Delta T = \frac{\rho_{T2}}{c}. \quad (5)$$

- 2) Calculate the satellite position at epoch T2 - ΔT using an interpolation algorithm and recalculate the distance between the satellite and the receiver.
- 3) Repeat steps (1) and (2) until the error between the two interpolated satellite positions is less than the threshold

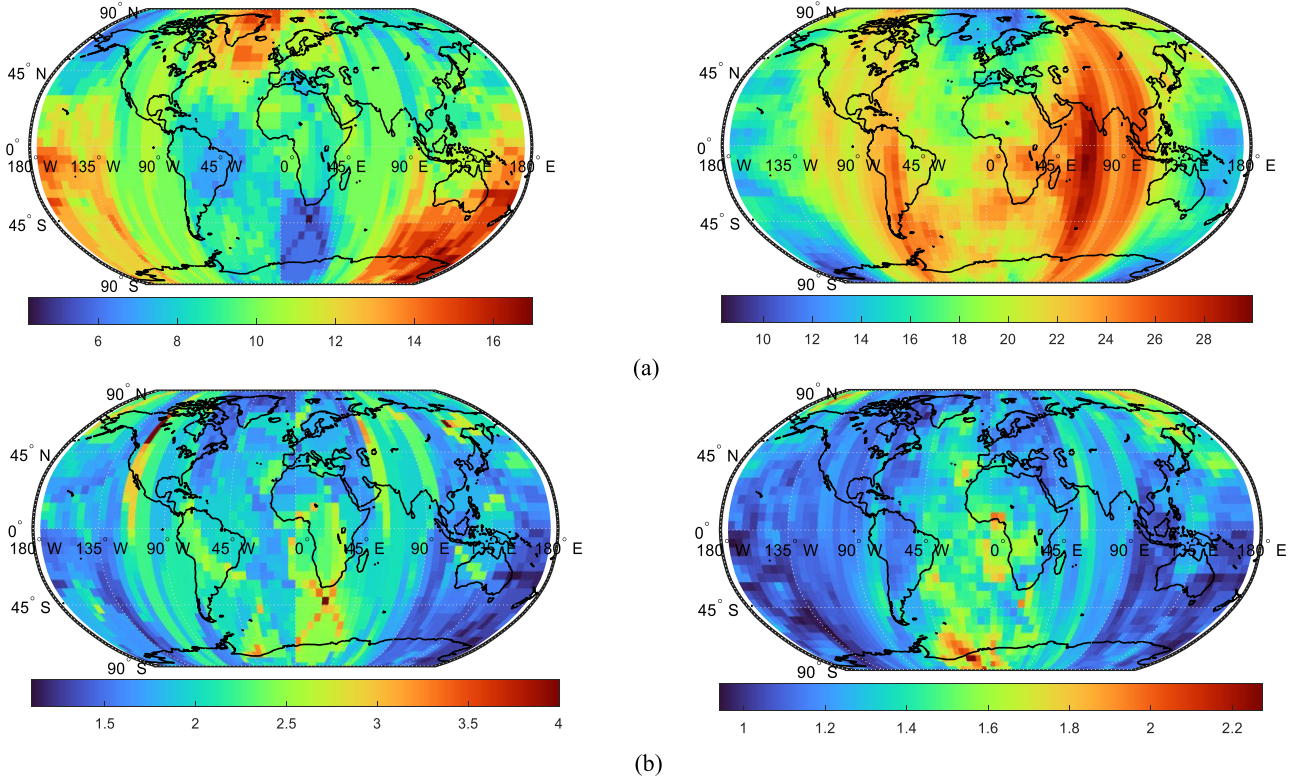


Fig. 6. Number of visible satellites and PDOP of BDS-3 globally before and after LEO satellites enhancement. (a) Number of visible satellites for BDS-3 only and BDS-3 enhanced with 244 LEO satellites. (b) PDOP of BDS-3 only and BDS-3 enhanced with 244 LEO satellites.

ε_T , and consider this to be the position of the LEO satellite at signal transmission epoch T_1 .

To initially verify the advantages of LEO satellite-enhanced GNSS, Fig. 6 shows the number of visible satellites and position dilution of precision (PDOP) of the BeiDou Navigation Satellite System Phase III (BDS-3) globally before and after LEO satellite enhancement. In the case of only the BDS-3 constellation, the maximum number of visible satellites is around 16, the minimum number is 5, and the maximum PDOP is 4. For integrity monitoring, the only BDS-3 may not meet the requirements in all situations. In the case of 244 LEO satellites enhancing BDS-3, the maximum number of visible satellites is around 28, the minimum number is 9, and the PDOP reaches a maximum of 2.2. For positioning solution and integrity monitoring, this will bring significant redundant observations and have important positive significance.

B. Masked CLA KalmanNet

Conventional KF is based on the assumption that the noise obeys a Gaussian distribution, but in the actual scene, due to the influence of MP/NLOS, the observation noise no longer obeys Gauss distribution, which will degrade the performance of the filter. KalmanNet is a method combining deep learning and traditional KF, which aims to solve the limitations of traditional KF in dealing with nonlinear dynamic model uncertainty [14]. By introducing an RNN module into the KF process, it learns to calculate Kalman gain (KG) from the data, so as to realize robust state estimation by learning the complex stochastic noise characteristics of the environment

while strictly adhering to the deterministic kinematic models of the navigation system.

Many existing studies are based on the LC KalmanNet framework, which features a simple structure and is easy to implement. However, it relies on position and velocity estimates independently obtained from GNSS, resulting in limited performance in environments with signal occlusion or stringent accuracy requirements. In contrast, the TC mode fuses data at the raw observation, offering higher accuracy and lower latency, and can more effectively suppress inertial navigation drift, thereby significantly enhancing overall system navigation reliability. To address these challenges, this article proposes a TC KalmanNet controlled by a masked matrix. Temporal alignment of data with asynchronous observation times is achieved via zero-padding. The masked matrix logic $\mathbf{M}$ is introduced to indicate whether the neural network should learn from specific inputs. In this section, we present the detailed algorithmic workflow of the masked KalmanNet. For the TC mode in the Earth-centered–Earth-fixed (ECEF) coordinate system, the state equation is formulated as follows:

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \boldsymbol{\omega} \quad (6)$$

where $\mathbf{x}$ represents the state vector, $\mathbf{F}$ denotes the system matrix, $\boldsymbol{\omega}$ represents the system noise, and the Gaussian noise assumption is relaxed. The state vector can be expressed as

$$\mathbf{x} = [\delta \mathbf{p} \quad \delta \mathbf{v} \quad \delta \boldsymbol{\theta} \quad \mathbf{b}_a \quad \mathbf{b}_g]^T \quad (7)$$

where $\delta \mathbf{p}$, $\delta \mathbf{v}$, and $\delta \boldsymbol{\theta}$, respectively, denote the position error, velocity error, and attitude error, while $\mathbf{b}_a$ and $\mathbf{b}_g$ represent

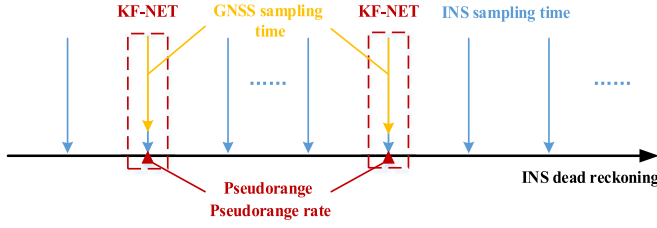


Fig. 7. Proposed GNSS/INS/LEO TC fusion system.

the deviation of the gyroscope and accelerometer, respectively. Based on the INS error propagation equations and the defined state variables, the detailed structure of the system matrix F can be found in [38], where comprehensive parameter definitions and derivations are provided.

Before employing the TC method, appropriate error compensation must be conducted. The standard compensation models for gyroscopes and accelerometers are outlined as follows:

$$\begin{aligned}\omega_{ib}^b &= (\mathbf{I} + \mathbf{S}_g + \mathbf{M}_g)^{-1} (\tilde{\omega}_{ib}^b - \tilde{\mathbf{b}}_g - \epsilon_g) \\ f_{ib}^b &= (\mathbf{I} + \mathbf{S}_a + \mathbf{M}_a)^{-1} (\tilde{f}_{ib}^b - \tilde{\mathbf{b}}_a - \epsilon_a)\end{aligned}\quad (8)$$

where $\tilde{\omega}_{ib}^b$ and $\tilde{f}_{ib}^b$ are the measured angular velocity and specific force, respectively; $\tilde{\mathbf{b}}_g$ and $\tilde{\mathbf{b}}_a$ are the estimated biases of the gyroscope and accelerometer, respectively; and ω_{ib}^b and f_{ib}^b are the bias-compensated true values used for navigation computation. The position relationship between GNSS and inertial measurement unit (IMU) in the integrated navigation system can be written as

$$\mathbf{Z} = \mathbf{p}_{SINS} + \mathbf{l}^e - \mathbf{p}_{GNSS} = \delta \mathbf{p} + \mathbf{R}_b^e \mathbf{l}^b \quad (9)$$

where $\mathbf{p}_{SINS}$ and $\mathbf{p}_{GNSS}$ are the positions of the IMU and GNSS receiver, respectively. $\mathbf{l}^b$ and $\mathbf{l}^e$ are the lever arm vectors in two different coordinate systems. Fig. 7 illustrates the proposed fusion system. The red box highlights the fusion timesteps, during which the IMU state is updated using GNSS/LEO observations. At other timesteps, the system relies on IMU propagation through recursive prediction.

1) *Data Preparation*: First, we list the features of the training input.

1) Observation innovation

$$\Delta \tilde{\mathbf{y}}_k = \mathbf{y}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k,k-1}. \quad (10)$$

2) Observation residual

$$\Delta \mathbf{y}_{k-1} = \mathbf{y}_{k-1} - \mathbf{H}_{k-1} \hat{\mathbf{x}}_{k-1,k-1}. \quad (11)$$

3) State innovation

$$\Delta \tilde{\mathbf{x}}_k = \hat{\mathbf{x}}_{k,k} - \hat{\mathbf{x}}_{k,k-1}. \quad (12)$$

4) State residual

$$\Delta \hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k,k} - \hat{\mathbf{x}}_{k-1,k-1}. \quad (13)$$

5) Residuals of the IMU at adjacent timesteps

$$[\Delta \alpha_k, \Delta \mathbf{w}_k] = [\alpha_k, \mathbf{w}_k] - [\alpha_{k-1}, \mathbf{w}_{k-1}]. \quad (14)$$

Since the dimensions of the state vector (including the innovation, residuals, and IMU outputs) remain constant over time, only the observed variables vary temporally. As a result, the input data at each timestep can be concatenated into a single vector

$$\mathbf{X}_k = [\Delta \alpha_k, \Delta \mathbf{w}_k, \Delta \hat{\mathbf{x}}_k, \Delta \tilde{\mathbf{x}}_k, \Delta \mathbf{y}_k, \Delta \tilde{\mathbf{y}}_k]. \quad (15)$$

To meet the input dimensional requirements of the neural network, zero-padding is applied to each input sequence based on the maximum number of observations N_{max} present in the dataset. N_k represents the number of pseudorange observations at epoch k . Correspondingly, a binary masked matrix $\mathbf{M}$ is constructed to differentiate valid observations from the zero-padded entries. This masked matrix ensures that the neural network does not mistakenly learn from artificially introduced zero-padding values

$$\bar{\mathbf{X}}_k = [\mathbf{X}_k, \mathbf{0}_{2N_{max}-2N_k}] \quad (16)$$

$$\mathbf{M}_k = [\mathbf{1}_k, \bar{\mathbf{0}}_{2N_{max}-2N_k}] \quad (17)$$

where $\bar{\mathbf{X}}_k$ and $\mathbf{M}_k$ are the training vector and corresponding masked vector at epoch k , respectively. $\mathbf{0}_{2N_{max}-2N_k}$ and $\bar{\mathbf{0}}_{2N_{max}-2N_k}$ represent the numerical vector and logical vector, respectively. T represents the number of epochs in the training dataset. Therefore, the training dataset is constructed as follows:

$$\mathbf{X} = [\bar{\mathbf{X}}_1, \bar{\mathbf{X}}_2, \dots, \bar{\mathbf{X}}_T] \quad (18)$$

$$\mathbf{M} = [\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_T] \quad (19)$$

$$\mathbf{Y} = [\bar{\mathbf{Y}}_1, \bar{\mathbf{Y}}_2, \dots, \bar{\mathbf{Y}}_T] \quad (20)$$

where $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{M}$ correspond to the input and output of the training dataset and the corresponding masked matrix, respectively. We fill each sequence with zeros to N_{max} . At the same time, the masked matrix $\mathbf{M}$ is constructed to ensure that CLA only depends on the real and effective timesteps and is not polluted by zero-padding

$$\mathbf{M}_{k,t} = \begin{cases} 1, & t < N_k \\ 0, & t \geq N_k. \end{cases} \quad (21)$$

2) *Masked CLA Network Structure*: Time convolution aggregates the features to capture local patterns with temporal dependencies. These local features are then passed to the LSTM layer, where contextual information with long-range dependencies is learned through inference. Finally, a self-attention mechanism is employed for the adaptive fusion of the feature information. The convolutional network with a masked matrix is illustrated

$$\hat{\mathbf{X}}_{k,t} = \mathbf{X}_{k,t} \cdot \mathbf{M}_{k,t}$$

$$\mathbf{Z}_{k,t} = \sum_{u \in \mathcal{W}(t)} \mathbf{W}_{o,\Delta(t,u)} \hat{\mathbf{X}}_{k,u}$$

$$\mathbf{S}_{k,t} = r \left(\frac{\mathbf{Z}_{k,t}}{\max(N_k, \epsilon)} + 1 [N_{k,t} > 0] \cdot \mathbf{b} \right) \quad (22)$$

where $\mathbf{W}_{o,\Delta(t,u)}$ is the convolution kernel weight, $\mathbf{b}$ is the bias vector, and $r(\cdot)$ represents the ReLU activation function. ϵ is a small constant introduced to prevent the occurrence of all-zero-padding. $\mathcal{W}(t)$ is the window index collection.

When zero-padding is applied across all windows, unbiased normalization implemented through an indicator function can effectively mitigate distortions arising from the insufficient number of valid samples near the boundaries. Furthermore, the feature S produced by the CNN, as well as the corresponding output masked matrix, should remain consistent with the input masked matrix to accurately reflect the validity of each step

$$\mathbf{M}_{k,t}^{\text{out}} = \mathbf{M}_{k,t}^{\text{in}}. \quad (23)$$

The masked matrix $\mathbf{M}_{k,t}^{\text{out}}$ will be passed to the subsequent LSTM and used to control which timesteps should be calculated. The input of the LSTM is the output S of the CNN network. At each timestep, the hidden state h_t and the memory cell c_t of the LSTM are updated under the guidance of the masked matrix. For positions where the mask is 0, the LSTM skips the update and retains the previous hidden and memory states. This allows the model to effectively ignore irrelevant timesteps while maintaining the temporal continuity of valid ones. The standard update equations for LSTM are given as follows:

$$\begin{aligned} i_k &= \sigma(\mathbf{W}_i s_k + \mathbf{U}_i h_{k-1} + \mathbf{b}_i) \\ f_k &= \sigma(\mathbf{W}_f s_k + \mathbf{U}_f h_{k-1} + \mathbf{b}_f) \\ o_k &= \sigma(\mathbf{W}_o s_k + \mathbf{U}_o h_{k-1} + \mathbf{b}_o) \\ \tilde{c}_k &= \tanh(\mathbf{W}_c s_k + \mathbf{U}_c h_{k-1} + \mathbf{b}_c) \\ c_k &= f_k \odot c_{k-1} + i_k \odot \tilde{c}_k \\ h_k &= o_k \odot \tanh(c_k). \end{aligned} \quad (24)$$

Under masked matrix control, the LSTM state updates at each timestep can be gated using the mask matrix $\mathbf{M}_{k,t}$

$$\begin{aligned} c_k^{(i)} &\leftarrow \mathbf{M}_{k,t} \cdot c_k^{(i)} + (1 - \mathbf{M}_{k,t}) \cdot c_{k-1}^{(i)} \\ h_k^{(i)} &\leftarrow \mathbf{M}_{k,t} \cdot h_k^{(i)} + (1 - \mathbf{M}_{k,t}) \cdot h_{k-1}^{(i)}. \end{aligned} \quad (25)$$

When updating the LSTM hidden states under normal conditions, the state is propagated step by step. However, if the current timestep corresponds to a padding position, the hidden state is typically retained from the previous timestep without update. In time-series modeling, not all timesteps contribute equally to the final prediction. Some moments contain discriminative features highly relevant to the prediction target, while others may provide only auxiliary information or even noise. Treating all timesteps uniformly dilutes the influence of critical features, ultimately degrading model performance. To address this, it is essential to introduce a mechanism that adaptively distinguishes the importance of different timesteps, enabling the model to focus on the most informative temporal components. The masked attention mechanism extends the standard self-attention structure by incorporating a masked matrix. This matrix regulates which positions in the sequence are eligible to contribute to the attention score computation, thus selectively attending to relevant information and improving the overall predictive capability of the model. When calculating the attention score, the current hidden state and the previous hidden state are used as inputs, where $e_{k,t}$ represents the attention score at epoch k

$$e_{k,t} = \mathbf{v}^T \tanh(\mathbf{W}_h \mathbf{h}_k + \mathbf{b}_h) \quad (26)$$

where $\mathbf{W}_h$, $\mathbf{b}_h$, and $\mathbf{v}$ are the learnable parameters. In this step, certain positions (denoted as $\mathbf{M}_{k,t} = 0$) are masked, meaning that they are ignored in the calculation by assigning large negative values to them. This ensures that the masked position does not affect the attention score. The formula for the attention score after masking is given as follows:

$$\tilde{e}_{k,t} = \begin{cases} e_{k,t}, & \text{if } \mathbf{M}_{k,t} = 1 \\ -\infty, & \text{if } \mathbf{M}_{k,t} = 0. \end{cases} \quad (27)$$

Attention weight α is calculated by applying a *softmax* function to the attention score, ensuring that the weights are normalized. This step determines how much attention the model should give to each input at each moment

$$\alpha_{k,i} = \frac{\exp(\tilde{e}_{k,i})}{\sum_j \exp(\tilde{e}_{k,j})} \quad (28)$$

the output vector is calculated by weighting and summing the input vectors h_i^k , and $\alpha_{k,i}$ is used to adjust the contribution of each h_i^k to the final output vector

$$c_k = \sum_i \alpha_{k,i} h_i^k. \quad (29)$$

The output of the attention mechanism outputs a KG matrix and inertial navigation measurement error estimation through the full connection layer controlled by a masked matrix. Then, the integrated navigation position is calculated. Fig. 8 shows the proposed masked CLA algorithm, with training outputs of the innovations and residuals shown in (10)–(14). Postprocessing position results as training labels. The neural network parameters in this article are optimized alternately [15]. It is important to distinguish the proposed masked matrix from the standard masking mechanisms found in Transformer architectures. In standard Transformers, masking primarily serves as a mathematical constraint for sequence modeling, acting as a task-agnostic utility to prevent information leakage. In contrast, our masked matrix is designed as a physical controller specifically for the time-varying dimensionality inherent in GNSS/LEO/INS integrated navigation. It addresses the dynamic fluctuation of visible satellites due to signal blockages. Unlike generic masking, which is often limited to the attention layer, our approach elevates the mask to a system-wide structural pillar that governs the entire feature extraction and fusion pipeline. We implement a masked adaptive CNN and LSTM. This goes beyond simple zero-padding suppression by normalizing kernels exclusively within the valid observation window, thereby eliminating spurious edge effects and ensuring the extraction of authentic features.

C. Training Criteria

KalmanNet is a supervised training network that can be trained end-to-end, and the training loss function is given as follows:

$$L = \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2 \quad (30)$$

where L is the loss function. $\mathbf{x}_k$ is the true value of the state at epoch k and can be obtained through postprocessing. We

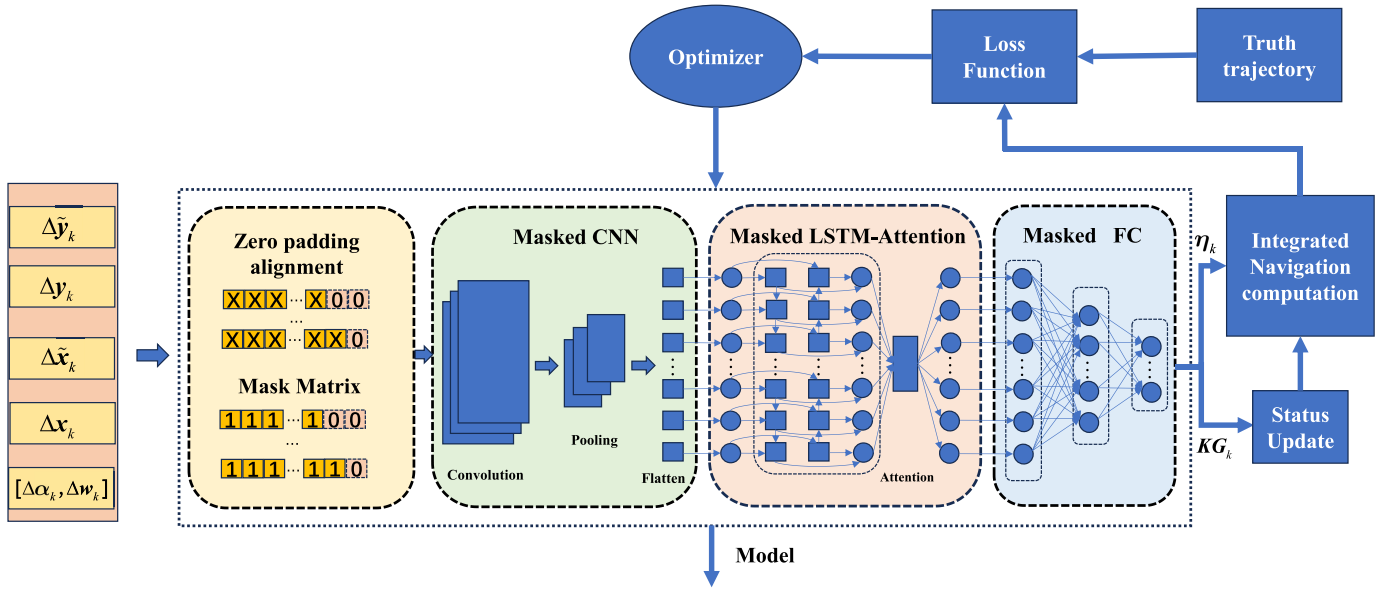


Fig. 8. Proposed masked CLA network.

use backpropagation to obtain K_k . The gradient function of the loss function on K_k is given as follows:

$$\begin{aligned} \frac{\partial L}{\partial K_k} &= \frac{\partial \|K_k \Delta x_k - \Delta y_k\|^2}{\partial K_k} \\ &= 2 \cdot (K_k \Delta x_k - \Delta y_k) \cdot \Delta x_k^T. \end{aligned} \quad (31)$$

Assuming the trainable parameter set Θ for the entire network (considering alternating optimization here [39]), including parameters from masked CLA, the regularization method is used to train the network based on the following criteria:

$$J(\Theta) = \frac{1}{T} \sum_{k=1}^T \|\hat{y}_k(x_k; \Theta) - y_k\|^2 + \gamma \cdot \|\Theta\|^2. \quad (32)$$

The parameter γ is the regularization coefficient. This method addresses the real-time state estimation problem in continuous-valued state-space models, focusing particularly on scenarios where noise statistics are unknown, and the system dynamics are only partially known or approximated via physical models. Traditional KF relies heavily on accurate knowledge of noise covariance matrices and domain-specific linear system modeling. In contrast, our algorithm replaces the analytical computation of the KG with a finite-complexity CLA integrated directly into the KF process, thereby bypassing the conventional requirement for matrix inversion and rigid statistical assumptions. This RNN-based KG estimator is trained in a supervised learning framework using labeled data, effectively learning to mimic the optimal KG behavior without explicit reliance on noise statistics. By learning the KG from data, our algorithm bypasses the need for statistical noise models and avoids the computational complexity associated with matrix inversion in traditional KF. Furthermore, it overcomes the need for custom adaptations required in extended KF (EKF) for nonlinear systems. Our algorithm maintains interpretability, exhibits low computational complexity, and

demonstrates high data efficiency, making it suitable for real-time applications. Our algorithm is grounded in the theoretical foundation of KF while selectively incorporating prior domain knowledge from the underlying SS model to enhance learning and inference. In summary, our algorithm represents an effective integration of model-based filtering theory with deep learning techniques, offering a scalable and flexible solution for real-time state estimation under uncertain or partially known conditions.

D. FDE Model

The enhancement of LEO satellites effectively compensates for the lack of visible GNSS satellites in complex environments, making it possible to conduct integrity monitoring. This article considers the relationship between statistical hypothesis testing and parameter estimation, and the fault detection formula is given as follows:

$$r_k = (r)^T (H_k P_k (H_k)^T + \sigma^2 I) r \quad (33)$$

where r stands for pseudorange residuals and pseudorange rate residuals. The detection threshold T_d can be calculated based on the false alarm probability. Once the fault is detected, the DIA formula is used to update the state vector and its covariance matrix

$$\begin{aligned} \hat{x}_i &= \hat{x}_0^+ - L_i \underline{y} \\ P_i &= P_0^+ + L_i Q_{vv} L_i^T. \end{aligned} \quad (34)$$

The detailed definition of L_i , Q_{vv} , and $\underline{y}$ can refer to [33], which is determined by different fault modes. Equation (34) represents the update of the state vector and covariance matrix after completing FDE. It is important to note that the FDE process is performed on the raw innovation sequence derived from the INS prediction. This ensures that the fault characteristics are preserved and detected before the KG, estimated by the neural network, is applied for state updating.

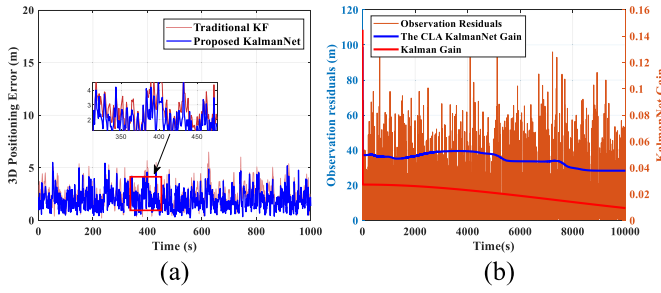


Fig. 9. Comparison of positioning error and KG between the two algorithms. (a) 3-D positioning errors of the two algorithms. (b) Relationship between observation residuals and Kalman gain norm.

Consequently, the proposed framework employs a dual-layer protection mechanism. The FDE module handles large, explicit faults by excluding them based on raw residuals, while the neural network provides a “soft exclusion” by downweighting observations with smaller errors that may escape the chi-square threshold.

E. Stability Analysis

Unlike traditional analytical methods that rely on Lyapunov stability proofs, our approach ensures stability implicitly through the end-to-end training process. To verify the stability of the proposed masked CLA network, this article simulates a complex environment by mixing 75% normal observations with 25% severe interference, constructing a highly challenging “thick-tailed” non-Gaussian noise model to stress test the robustness of navigation algorithms under frequent and large-amplitude error interference. As shown in Fig. 9, the comparative performance analysis between the proposed CAL KalmanNet and the traditional KF is illustrated, evaluating both positioning accuracy and gain adaptivity. As depicted in Fig. 9(a), the 3-D positioning error of KalmanNet is consistently lower than that of the traditional KF throughout the 1000-s simulation. This performance gain is further elucidated in Fig. 9(b), which characterizes the evolution of observation residuals and KGs over a 10 000-s duration. Unlike the traditional KG, which exhibits a monotonic decay regardless of measurement fluctuations, the CLA KalmanNet gain demonstrates a distinct adaptive behavior in response to the stochastic nature of the observation residuals. By dynamically modulating the gain based on real-time measurement quality, the CLA KalmanNet effectively compensates for model mismatches and nonstationary noise, thereby achieving a more precise and reliable state estimation than conventional model-based approaches.

Spectral radius is an important criterion for system stability [40]. Fig. 10 illustrates the evolution of the spectral radius for both the traditional KF and the proposed CLA KalmanNet over 10 000 testing epochs. This metric serves as a critical indicator of the asymptotic stability and convergence characteristics of the filtering systems. As demonstrated in the numerical results, while both algorithms exhibit an initial transient increase in spectral radius, they both successfully converge to steady-state values strictly within the unit circle. This convergence confirms the asymptotic stability of the closed-loop estimation

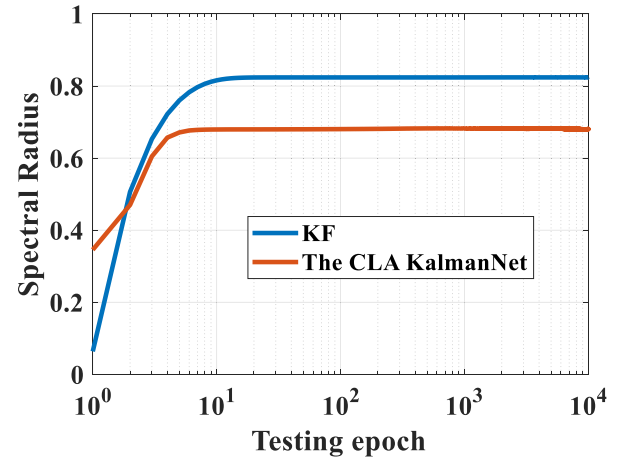


Fig. 10. Comparison of spectral radii of the gain matrices of the two algorithms.

TABLE I
PARAMETERS AND ENVIRONMENTS OF THE EXPERIMENTAL DATASETS

Dataset	Description	Dynamic Type	Duration
Training Dataset	High building occlusions, frequent MP/NLOS.	Kinematic	About 150 min
Testing Dataset	Overpasses and tree-lined roads	Kinematic	About 110 min

processes for both methods. Notably, the CLA KalmanNet achieves a steady-state spectral radius of approximately 0.68, which is significantly lower than the 0.82 observed for the traditional KF.

From a control-theoretic perspective, the reduced spectral radius of the proposed masked CLA KalmanNet implies a faster convergence rate and a superior stability margin. These results suggest that by maintaining a smaller spectral radius, the CLA KalmanNet provides enhanced robustness and numerical stability against nonstationary noise and model uncertainties, further validating its reliability for long-term state estimation tasks.

III. EXPERIMENTAL VERIFICATION

This experiment is conducted in two stages, utilizing two real-world datasets [41], with one designated for training and the other for testing. Both datasets are collected in complex environments, including signal-obstructed environments such as overpasses, tree-lined roads, and urban buildings (see Table I). The dataset can be obtained from the following network address: <https://www.kaggle.com/datasets/fengzhusgg/smartpnt-pos>. In the absence of actual LEO satellite measurements, the approach described in Section II is employed to generate simulated LEO satellite downlink data. This section demonstrates how to achieve joint position calculation using real GNSS data and LEO simulation data through coordinate transformation and timestamp alignment.

TABLE II
EXPERIMENTAL PARAMETERS IN THIS ARTICLE

Fusion source types	Configuration parameter	
IMU	Accelerometer	BS(mg) 0.01
		WH($m/s/\sqrt{hr}$) 0.03
	Gyroscope	BS(deg/hr) 0.01
		WH($deg/\sqrt{hr}$) 0.03
GNSS	Global Positioning System (GPS)	32 MEO
	BDS-3	24 MEO+3 GEO+3 IGSO
LEO	The proposed mixed constellation	144 Walker+81 Polar

A. Experimental Parameter Settings

The relevant experimental parameters are listed in Table II. The constellation configuration of LEO satellites is primarily categorized into the walker and polar constellations. The walker constellation offers excellent coverage at low latitudes, whereas the polar constellation provides superior performance in high-latitude regions [42]. To enhance GNSS performance further, this study simulates a hybrid constellation that combines the advantages of both. The IMU error model parameters are presented in Table II, including quantization white noise, bias instability, and random walk. In addition, the abbreviations “BS” and “WH” denote the zero-bias instability and white noise components of the IMU, respectively. First, we use two real-world dynamic datasets, containing real GNSS observations, real satellite ephemeris, and clock errors. The real ground trajectory is obtained through postprocessing, with a positioning accuracy of 0.1 m.

Second, the simulated LEO data observations are rigorously aligned to this realistic spatiotemporal framework. The LEO satellite orbits are generated and interpolated based on the exact GPS time tags recorded by the real-world receiver, ensuring strictly synchronous processing without temporal misalignment. Both the simulated LEO constellation and real GNSS measurements are unified into the WGS-84 coordinate frame. The error model is derived from real data. It is important to acknowledge that the height-dependent linear interpolation model used for LEO ionospheric delay is a simplification of the complex, stochastic fluctuations found in real-world total electron content (TEC) distributions. While this model may underestimate certain high-order spatiotemporal inhomogeneities, it provides a consistent and controlled benchmark where all comparative algorithms are subjected to identical error profiles.

Furthermore, we contend that this simplification likely yields a conservative estimate of the proposed method’s superiority. Traditional filters based on Gaussian assumptions typically struggle with complex, nonstationary ionospheric errors. In contrast, data-driven approaches like the masked CLA network excel at learning and compensating for such nonlinear patterns. As shown in Fig. 5, our modeling of



Fig. 11. Training trajectory of the vehicle.

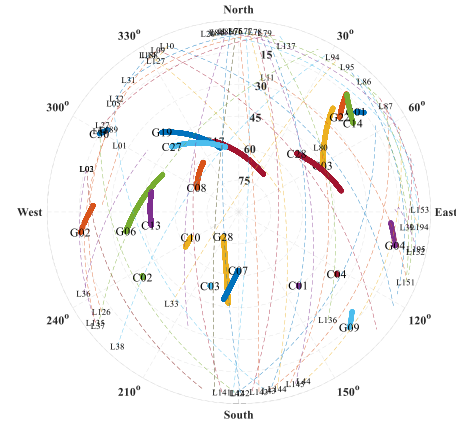


Fig. 12. Visible satellite sky-plot of the training dataset.

LEO MP/NLOS error and observation noise does not employ Gaussian models. Instead, we statistically analyze real vehicle-mounted GNSS data to obtain its non-Gaussian, heavy-tailed, and complex characteristics related to satellite elevation angle and C/N0, and inject these statistical characteristics into the LEO simulation observations. Therefore, the goal of network learning is to predict the optimal state correction from these real-world measurements. This is more challenging and realistic than training in a completely ideal simulation environment.

B. Analysis of Training Dataset Results

This section primarily describes the training dataset, which is used to train the CLA network to learn network parameters. These parameters will then be applied to the test dataset in Section III-C to evaluate the effectiveness of the proposed algorithm. The training platform is based on an Intel Core i7-12500H 64-bit Windows 11 system. Fig. 11 shows the training trajectory of the vehicle.

Fig. 12 presents the observed satellite sky-plot for the training dataset. The observation environments are notably complex, encompassing challenging scenarios such as urban canyons and tree-lined roads. These conditions significantly reduce the number of observable GNSS satellites, thereby limiting the effectiveness of integrity monitoring. However, with the integration of LEO satellite observations, a substantial increase in the number of available satellite signals is achieved. The masking angle of LEO satellites is simulated using a random model [43].

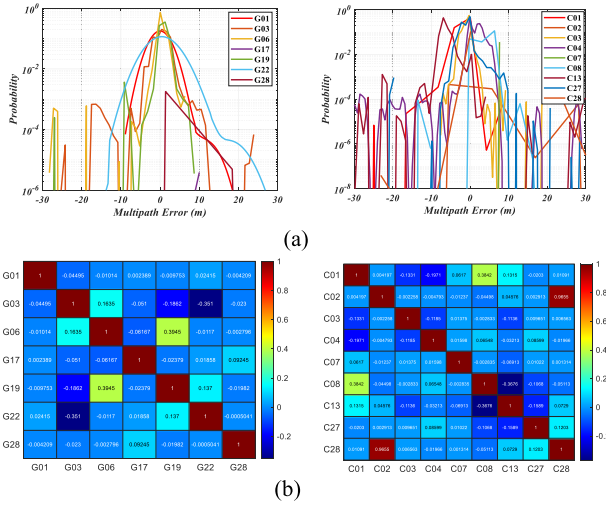


Fig. 13. Probability density function and correlation heatmaps of MP/NLOS errors of different satellites. (a) Satellite MP/NLOS errors distribution of the training dataset. (b) Satellite MP/NLOS errors correlation heatmaps of the training dataset.

Fig. 13 depicts the probability density function (pdf) and correlation heatmaps of MP/NLOS errors for both GPS and BDS-3 in the training dataset. It is evident that, due to the complex observational environment, the MP/NLOS error distributions for both constellations exhibit pronounced non-Gaussian characteristics, thereby violating the Gaussian noise assumption commonly adopted in traditional KF. Applying such filters without adaptation can result in significant degradation in navigation and positioning performance. Moreover, the presence of heavy-tailed error distributions may lead to underestimation of protection levels (PLs), thereby elevating the integrity risk and reducing the overall usability of the algorithm. As shown in Fig. 13(b), the pseudorange residuals among different satellites within the GPS and BDS-3 constellations exhibit a noticeable degree of correlation. This intersatellite correlation arises primarily from shared error sources, including satellite clock biases, atmospheric delays, observation geometry, data processing techniques, and hardware-induced deviations. Properly accounting for these correlations is essential for optimizing positioning algorithms, especially in the formulation of weighting strategies and fault detection mechanisms, and is fundamental to enhancing the robustness of integrity monitoring systems. However, conventional integrity monitoring methods often assume statistical independence among pseudorange residuals. In the presence of correlation, this assumption may lead to an elevated false alarm rate. Correlated error sources can affect multiple satellite observations simultaneously, thereby impairing the system's ability to isolate and identify faulty measurements. Consequently, such correlated errors may be misclassified as nominal or low risk, posing a threat to the overall safety of the navigation system. PLs, widely used in integrity monitoring, are statistical bounds intended to assess the reliability and accuracy of positioning solutions. Their computation depends critically on the statistical properties of the observation residuals. If these residuals exhibit significant correlation, the

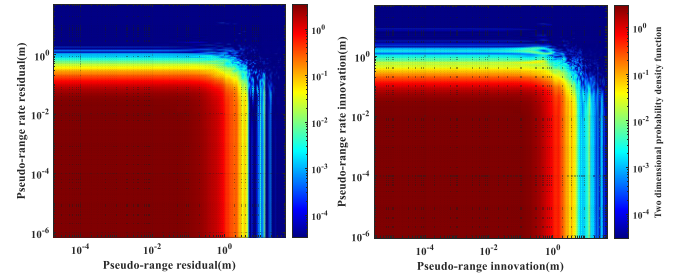


Fig. 14. Joint distribution of residuals and innovation of pseudorange and pseudorange rate in the training dataset.

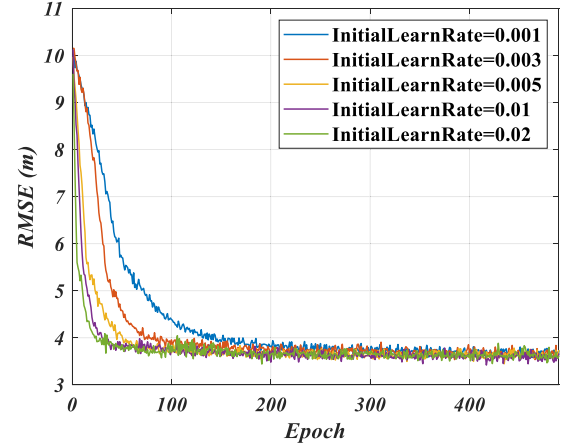


Fig. 15. Learning curves at different initial learning rates.

TABLE III
TRAINING NETWORK CONFIGURATION

Name	Additional information
Conv1D	24 filters of size(3 x 3)
	stride(1 x 1)
LSTM	ReLU
	64 units,5 hidden layers
	0.2 dropout
Loss function	Mean Square Error (MSE)
Optimizer	Adam
Initial Learning Rate	0.01

monitoring system may underestimate the true uncertainty, leading to misestimated PLs.

Fig. 14 presents the innovations and residuals of the pseudorange and pseudorange rate observations from the training dataset. As shown in Fig. 14, both innovation and residuals deviate from Gaussian distributions due to MP/NLOS caused by the observation environment. In particular, the pseudorange innovation and residual exhibit more complex multimodal distributions.

Table III summarizes the detailed parameter configuration used for training, including those associated with the masked CLA network. The loss function adopted in this article is the average variance of prediction errors, and the Adam optimizer is employed for network training due to its adaptive

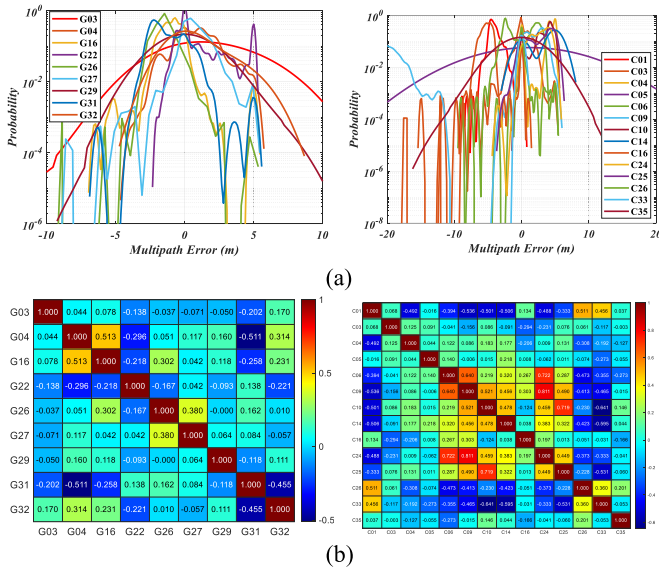


Fig. 16. Probability density function and correlation heatmaps of MP/NLOS errors of different satellites. (a) Satellite MP/NLOS error distribution of the testing dataset. (b) Satellite MP/NLOS error correlation heatmaps of the testing dataset.

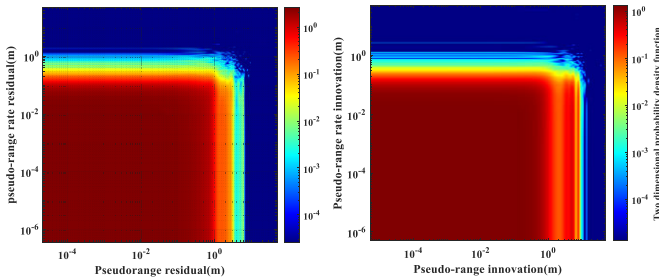


Fig. 17. Joint distribution of pseudorange and pseudorange rate innovation and residual in the testing dataset.

learning rate capabilities and computational efficiency. Fig. 15 illustrates the learning curves obtained under various initial learning rates. As observed, a higher initial learning rate leads to faster convergence during the early training stages. However, when the initial learning rate is increased to 0.02, the post-convergence oscillations become significantly more pronounced, indicating reduced training stability. Taking both convergence speed and stability into consideration, an initial learning rate of 0.01 is selected as the optimal setting for this article.

C. Analysis of Testing Dataset Results

This section evaluates the positioning performance and integrity monitoring capabilities of the proposed algorithm using the testing dataset, providing empirical validation of its effectiveness. The analysis begins by characterizing the stochastic properties of the dataset, with a focus on MP/NLOS errors across different observation types and their intersatellite correlations. Subsequently, the joint distribution of the KF residuals and innovations is investigated. Finally, the proposed method is benchmarked against existing algorithms in terms of positioning accuracy and integrity PLs.

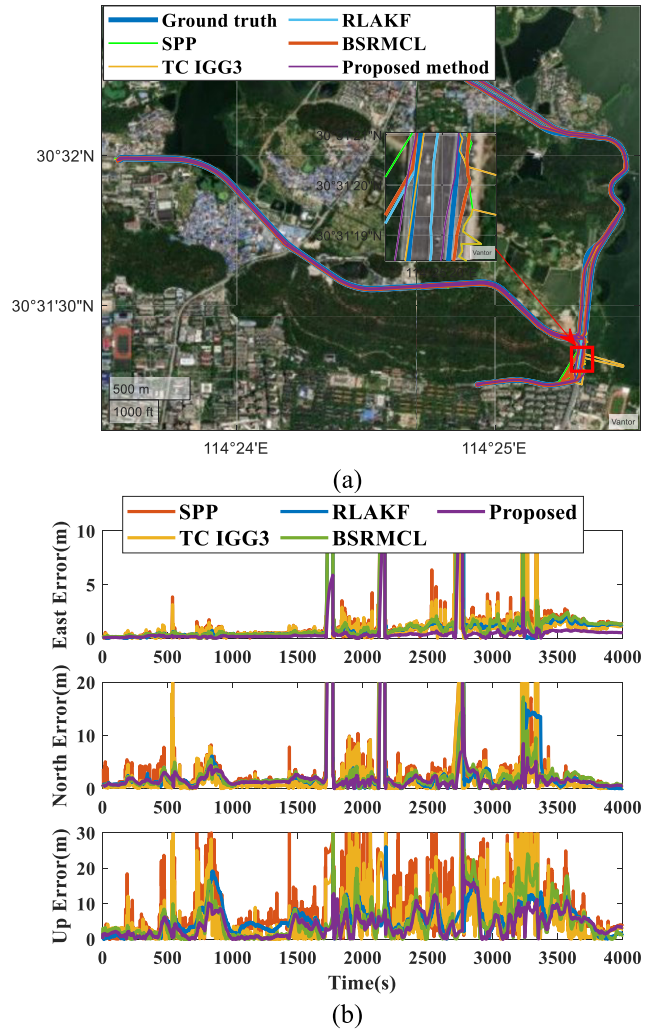


Fig. 18. Comparison of 2-D trajectories and positioning errors in the east, north, and down directions of different algorithms in the testing dataset. (a) Two-dimensional trajectories of different algorithms. (b) Positioning errors in three directions of different algorithms.

Fig. 16(a) shows the pdf of the MP/NLOS errors of GPS and BeiDou-3 based on the test dataset. Clearly, due to the complex observation environment, the MP/NLOS errors of both constellations exhibit a non-Gaussian distribution. This observation is consistent with the training dataset and contradicts the Gaussian noise assumption commonly used in traditional KF. As shown in Fig. 16(b), the pseudorange residuals of different satellites within the GPS and BDS-3 constellations show a significant correlation.

Fig. 17 presents the joint distribution of Kalman residuals and innovations from the testing dataset. As shown in Fig. 17, both sets of data deviate from Gaussian distributions due to MP/NLOS errors caused by the observation environment. In particular, the pseudorange innovations and residuals also exhibit more complex multimodal distributions.

This article selects five existing positioning algorithms for comparative verification in a testing dataset.

- 1) *SPP*: The SPP methods.
- 2) *TC IGG3* [44]: A robust IGG3 algorithm used in GNSS/INS TC mode.

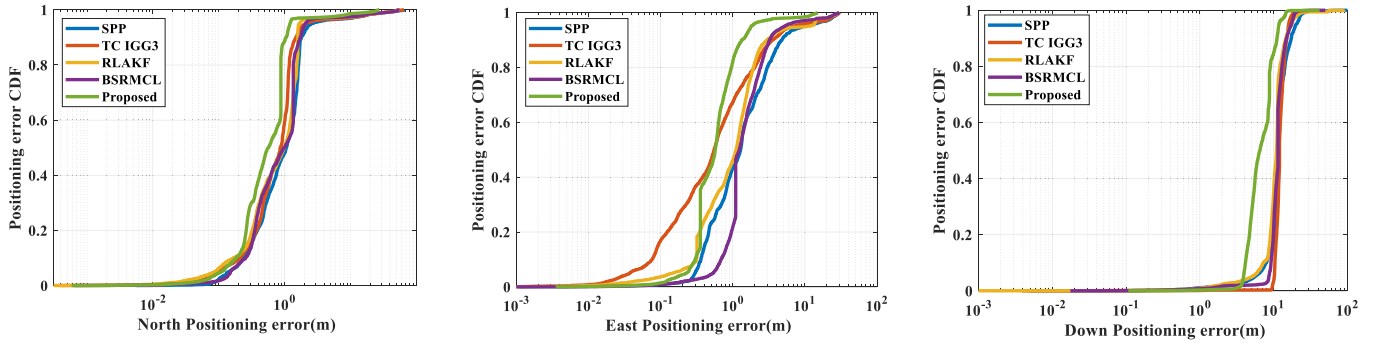


Fig. 19. Comparison of positioning errors in the east, north, and down directions in the testing dataset using different algorithms.

- 3) *BSRMCL* [24]: A CNN–LSTM-assisted GNSS/INS integrated navigation system that utilizes a bias and scale factor recomputed method (BS-RM) to predict IMU errors, thereby maintaining consistent positioning during GNSS outages.
- 4) *RLAKF* [45]: An adaptive KF navigation algorithm based on reinforcement learning.
- 5) Masked CLA methods proposed in this article.

Among the algorithms mentioned above, SPP and TC IGG3 are model-driven methods, RLAKF and BSRM are data-driven methods using neural networks, and our proposed masked CLA method combines both data and model aspects. Fig. 18 comprehensively compares the proposed algorithm with existing algorithms such as RLAKF, BSRMCL, and traditional SPP and TC IGG3 in terms of 2-D trajectory tracking and 3-D localization errors on the test road section. From the 2-D trajectory visualization in Fig. 18(a), the algorithms exhibit clear performance stratification. Traditional SPP and TC IGG3 algorithms show severe trajectory divergence on complex road sections. In contrast, RLAKF and BSRMCL, as existing robust filtering methods, exhibit significantly improved overall performance, enabling them to follow the ground truth trajectory accurately on a large scale. However, in locally magnified detailed areas, these two methods still fail to completely overcome environmental interference; RLAKF and BSRMCL both exhibit varying degrees of trajectory drift and deviation from the ground truth. Only the proposed method maintains a high degree of overlap with the ground truth in such challenging areas, demonstrating its localization advantage in solving localized complex scenarios. The 3-D error series in Fig. 18(b) further quantifies this difference. While RLAKF and BSRMCL effectively reduced the massive errors of tens of meters inherent in the SPP algorithm, both algorithms still exhibited instability throughout the 4000-s test period. Particularly in the high-dynamic range of 1700–3500 s, RLAKF still showed significant abrupt peaks in the north and east directions, while BSRMCL exhibited persistent high-frequency jitter and residual errors of several meters in the up direction. In stark contrast, the proposed method maintained stable convergence of the error curves even when RLAKF and BSRMCL showed significant fluctuations, with positioning errors in all three directions consistently kept at extremely low levels. This clearly demonstrates that, compared to the existing RLAKF and BSRMCL algorithms, the

proposed method offers a decisive improvement in suppressing anomalous observations, resisting MP/NLOS interference, and enhancing the overall robustness of the system.

Fig. 19 uses the cumulative distribution function (cdf) to statistically analyze the positioning error distribution in the north, east, and down directions of the testing dataset, further verifying the performance differences between different algorithms from a probabilistic statistical perspective. The results show that the proposed method is located on the far left of SPP, TC IGG3, and the existing RLAKF and BSRMCL algorithms in all three dimensions, exhibiting the steepest upward trend, indicating that the algorithm can achieve a higher confidence level within a significantly smaller error threshold.

In the horizontal direction, the proposed method converges to a cumulative probability of 1.0 earlier than RLAKF and BSRMCL, effectively overcoming the long-tailed error problem of traditional methods. Its advantage is particularly prominent in the down direction, while the error curves of the other four comparative algorithms are mainly concentrated in the range above 10 m, the proposed method has already converged within a range of several meters, forming a significant performance gap with other algorithms. These statistical results strongly demonstrate that the algorithm presented in this article not only performs well in local road sections but also surpasses existing robust filtering algorithms in terms of overall error distribution. In particular, it demonstrates excellent robustness and reliability in solving the common problem of inaccurate GNSS elevation positioning.

Table IV presents the RMSE statistics for the SPP, TC IGG3, RLAKF, and BSRMCL algorithms alongside the proposed method across north, east, down, and 3-D dimensions. The results indicate that traditional SPP and TC IGG3 methods suffer from limited accuracy, exhibiting 3-D RMSEs of 15.40 and 14.36 m, respectively. Although the robust RLAKF and BSRMCL algorithms mitigate these errors, lowering 3-D RMSE to 10.90 and 9.83 m, the proposed method demonstrates the most significant performance gains. Specifically, it achieves north, east, and down RMSEs of 3.71, 2.72, and 5.18 m, respectively, culminating in a minimal 3-D RMSE of 6.93 m. Improvement analysis reveals that the proposed approach enhances 3-D positioning accuracy by 55.00% over SPP, including a remarkable 60.58% reduction

TABLE IV
COMPARISON OF POSITIONING PERFORMANCE OF DIFFERENT POSITIONING ALGORITHMS IN THE TESTING DATASET

Algorithm	Position RMSE (m)			
	North	East	Down	3D
SPP	5.33	6.00	13.14	15.40
TC IGG3	4.89	5.64	12.27	14.36
RLAKF	4.64	5.37	8.27	10.90
BSRMCL	4.28	5.42	6.99	9.83
Proposed	3.71	2.72	5.18	6.93
Improvement (compared with SPP)	30.39%	54.67%	60.58%	55.00%
Improvement (compared with TC IGG3)	24.13%	51.77%	57.78%	51.74%
Improvement (compared with RLAKF)	20.04%	49.35%	37.36%	36.42%
Improvement (compared with BSRMCL)	13.32%	49.82%	25.89%	29.50%

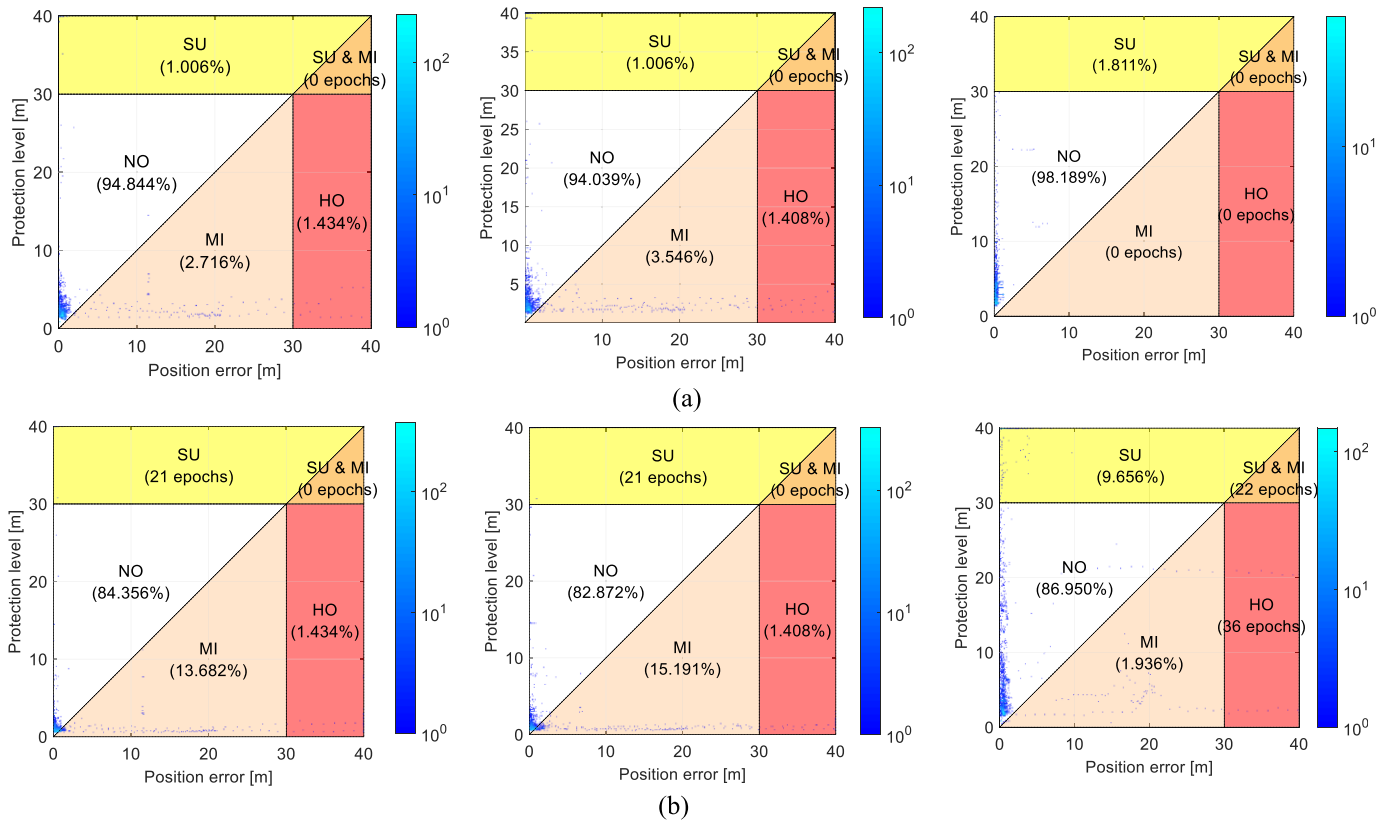


Fig. 20. Stanford diagram of different algorithms using the testing datasets (RAIM, RANSAC, and proposed). (a) Vertical direction. (b) Horizontal direction.

in vertical error, effectively addressing GNSS elevation weaknesses. Furthermore, compared to the BSRMCL method, the proposed algorithm secures a 29.50% improvement in 3-D accuracy, with the east component improving by 49.82%. These statistics underscore the proposed method's superior error suppression capabilities and robustness in complex environments.

To evaluate the integrity monitoring performance of the proposed framework on the testing dataset, we utilize Stanford diagram to visualize integrity events and assess overall system reliability. In these diagrams, the positioning error and PLs for each epoch are plotted on the horizontal and vertical axes, respectively, with separate diagrams provided for the horizontal and vertical components. In addition, the

diagrams assess system availability, indicated by the region where the PLs remain below the alert limit (AL). A color gradient is employed to depict the density distribution of the scatter points. Fig. 20 presents Stanford diagrams for the proposed algorithm alongside the RAIM and RANSAC benchmarks. Evidently, the proposed method demonstrates a significantly higher probability of operating within the nominal safety region. In the vertical domain, our algorithm achieves a 98.19% probability of safe operation, outperforming RAIM (94.84%) and RANSAC (94.04%). Similarly, in the horizontal direction, our algorithm reaches 86.95%, compared to 84.36% for RAIM and 82.87% for RANSAC. Consequently, the proposed algorithm exhibits superior integrity monitoring capabilities on the testing dataset. By substantially reducing

the number of epochs where the system is unavailable, our approach significantly enhances overall system availability in both the horizontal and vertical directions.

IV. CONCLUSION

This article proposes a CLA network augmented by a masked matrix to facilitate the implementation of the GNSS/INS/LEO TC algorithm within a KalmanNet architecture for complex environments. Initially, a comprehensive simulation for generating LEO downlink observations is established. To estimate the KG within the TC framework, a CLA network regulated by a masked matrix is employed, incorporating a zero-padding strategy to accommodate the variable dimensionality of measurement vectors. Crucially, the masked matrix enables the network to selectively extract valid information, thereby enhancing robust performance. Furthermore, this work integrates navigation parameter estimation with FDE to guarantee solution stability and reliability. Experimental results demonstrate that the proposed method outperforms SPP, TC IGG3, RLAKF, and BSRMCL, achieving accuracy improvements of 55.53%, 51.74%, 36.42%, and 29.50%, respectively. The superiority of the proposed approach in integrity monitoring is further corroborated by Stanford diagrams. Despite these contributions, certain limitations remain. To further validate the proposed method in real-world scenarios, future work will involve real LEO observation data. Specifically, we plan to utilize publicly available navigation data from existing LEO constellations (such as the CentiSpace or LuoJia-series satellites). In addition, data support will be sought through collaborative research projects involving LEO-based enhancement systems to ensure the practical applicability and foresight of the research.

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